

parameter $\lambda \rightarrow \infty$, we recover a Walrasian interbank market. In a Walrasian market, if the banking system has an overall excess of reserves, we have $\bar{i}_t^f = i_t^m$, while if the banking system has an overall deficit of reserves, we have $\bar{i}_t^f = i_t^w$. Meanwhile, if aggregate excess reserves are exactly zero, the Fed funds rate is indeterminate. This implies that the costs of deficits equals the benefits of a surplus, $\chi_t^+ = \chi_t^-$ and changes in withdrawal risks would have no effects. In addition, OMO would be neutral unless aggregate excess reserves change sign, for example, in a Walrasian market with $\rho = 0$, there are no effects of OMO because aggregate excess reserves are always positive in this case.

Figure 1 presents χ^+ , χ^- and the Fed funds rate as a function of the log inverse of market tightness θ , for the frictional OTC market (left panel) and the Walrasian market (right panel). In the case of the OTC market, we can see how as liquidity increases and we move along the x-axis, the Fed funds rate falls closer to the floor of the corridor. The figure illustrates how depending on the target for the interbank market rate, the central bank can adjust the amount of liquidity to aim at a desired target.

The classic [Poole](#) model also generates a smooth downward curve for the interbank market rate as a function of the real supply of government liquidity, as in Figure 1(a). However, it does so by assuming that the interbank market, modeled as a Walrasian market, closes before withdrawal shocks are realized. Like [Afonso and Lagos \(2015\)](#), our model can thus be seen as a microfoundation of such a downward-sloping relationship.²² Notably, the model predicts that withdrawal risk can have very different implications depending on the interbank market's functioning. Another notable difference is that the Poole framework is, in effect, a partial equilibrium model and, therefore, does not allow for a joint analysis of prices, credit, and macroeconomic aggregates. When we present the model's applications in the next section, we will show how embedding this OTC interbank market in a general equilibrium model gives rise to novel policy implications regarding

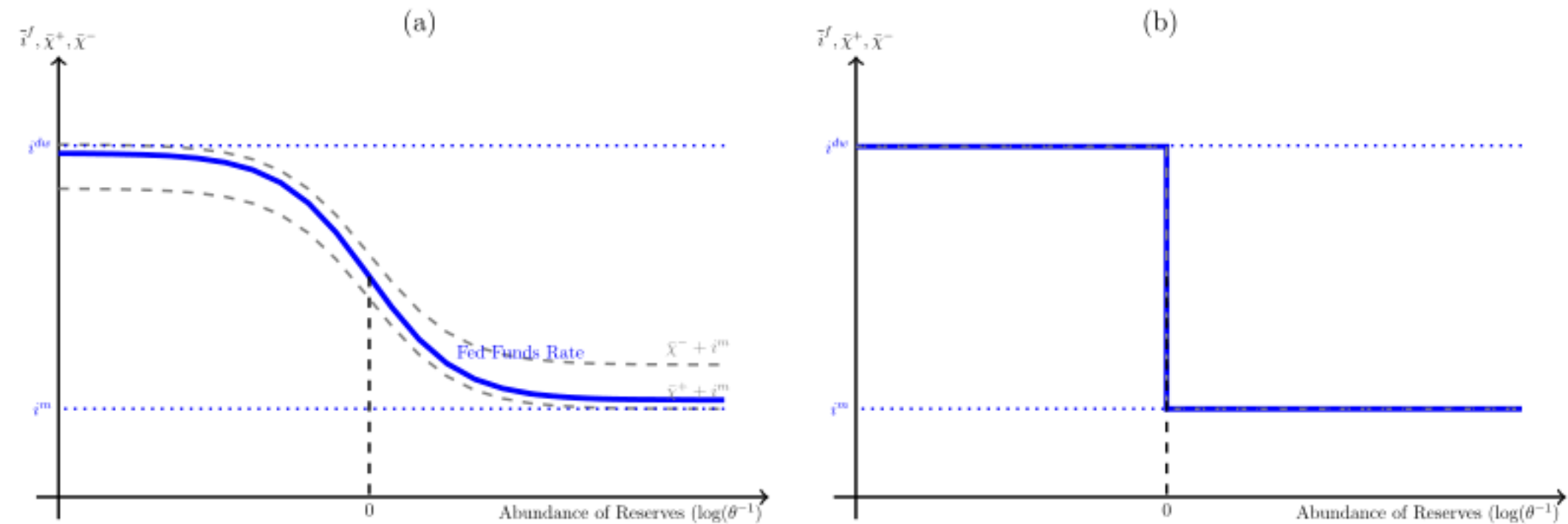


FIGURE 1.—OTC versus Walrasian markets.

²²See [Bianchi and Bigio \(2017\)](#) for a derivation